

Title

Local AI Data Analyst Stack
n8n + Ollama + Qwen3 + Supabase
Step-by-Step Installation & Demo
Guide

This is a complete Local AI Data Analyst System running on a laptop without OpenAI, Claude, Gemini, OpenRouter, API keys or token charges .

Ollama (Local Models) is the closest thing to "unlimited."

You run models on your own computer:

- No token billing
- No API charges
- Limited only by your hardware (RAM, GPU, CPU)

Examples:-

- Llama 3
- Qwen
- Mistral

If your PC can run the model, you can generate millions of tokens without paying per token

What are Tokens?

A **token** is the unit that AI models use to process text.

A token is not exactly a word:

Text	Approximate Tokens
Hello	1 token
Hello, how are you?	5–6 tokens
100 words of English	~130 tokens
1 page of text	~500–800 tokens

When you interact with an LLM, both your input and the AI's response consume tokens.

1. Free Models

Some open-source models may be available for free or with promotional free usage.

Examples can change over time:

- Certain Meta Llama models
- Some Mistral models
- Community-hosted models

These typically have:

- Rate limits
- Daily limits
- Availability limits

So "free" does **not** mean unlimited.

Ollama (Local Models)

[Ollama](#) ↗

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Prerequisites

- Hardware:
 - Windows 11
 - Intel i5-1135G7 (or similar)
 - 16 GB RAM recommended
 - 20+ GB free space for models
- Accounts:
 - Supabase account
 - GitHub access (for workflow JSON)
- Software:
 - Node.js
 - Ollama
 - n8n

Local LLM vs Online LLM

Feature	Local LLM (Ollama + Qwen)	Online LLM (OpenAI/OpenRouter/Gemini)
Internet Required	✗ No	✓ Yes
API Key Required	✗ No	✓ Yes
Usage Charges	✗ Usually free after download	✓ Pay per token/request
Privacy	✓ Data stays on PC	⚠ Data sent to provider
Speed	Depends on your hardware	Usually faster
Model Quality	Good	Usually better
Maintenance	You manage it	Provider manages it
Offline Usage	✓ Yes	✗ No
Scalability	Limited by your PC	Massive
Data Security	High	Depends on provider

Step 1 - Install Node.js

- 1. Download Node.js LTS from nodejs.org
- 2. Run installer with default options
- 3. Open Command Prompt

Verify:

- `node -v`
- `npm -v`

Example:

- v24.16.0
- 11.13.0

Step 9: Install Node.js

1. Go to:

[Node.js Official Downloads](#)

2. Download the **LTS (Long Term Support)** version for Windows.

3. Run the installer.

- Keep all default options.
- Make sure **"Add to PATH"** remains checked (it is checked by default).

4. When installation finishes:

- Close all Command Prompt windows.
- Open a **new** Command Prompt.

5. Run:

```
</> Bash
```

```
node -v
```

You should see something like:

```
v22.x.x
```

Step 10: Verify Node Directly

In Command Prompt, run:

```
</> cmd
```

```
"C:\Program Files\nodejs\node.exe" -v
```

Step 12: Verify npm

In the same Command Prompt, run:

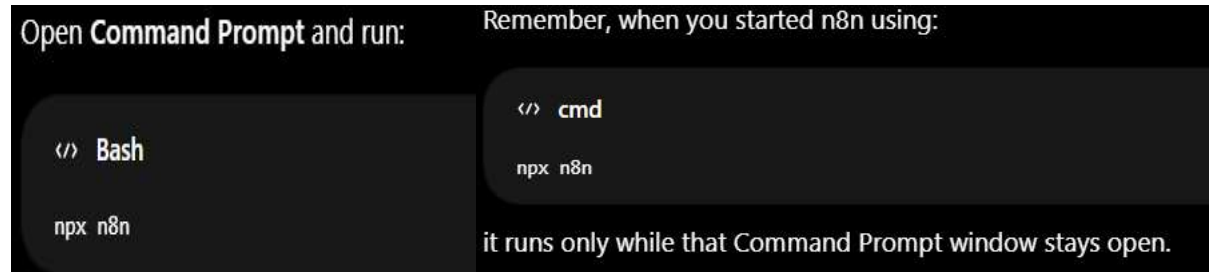
```
</> cmd
```

```
npm -v
```

Step 2 - Install and Start n8n

- Open Command Prompt:

- `npx n8n`



- Expected:
- Editor is now accessible via:
- `http://localhost:5678`
- Create Owner Account when prompted.

Step 3 - Install Ollama

- Download Ollama from ollama.com

Verify installation:

- `ollama --version`

Start Ollama service automatically after install.

1. Visit:

[Ollama Official Website](https://ollama.com) ↗

2. Click **Download** and choose **Windows**.

3. Run the installer and complete the installation using the default options.

4. After installation finishes, open **Command Prompt**.

5. Run:

```
</> Bash
```

```
ollama --version
```


Step 4 - Download Models

- Download models:
ollama pull qwen3:4b
ollama pull qwen3:8b
- Verify:
- ollama list

Step 3: Download Your First AI Model

We'll start with Qwen3 4B, which should run well on your laptop.

Open **Command Prompt** and run:

```
</> Bash  
ollama pull qwen3:4b
```

Then run:

```
</> Bash  
ollama list
```

You should see something similar to:

NAME	ID	SIZE
qwen3:4b	xxxx	x.x GB

Step 5 - Test Ollama

- Run model:
- `ollama run qwen3:4b`
- JSON showing installed model

Step 4: Test the Model Locally

In Command Prompt, run:

```
</> Bash
```

```
ollama run qwen3:4b
```

When the prompt appears, type:

```
What is PostgreSQL? Explain in 3 sentences.
```

The AI should generate a response.

Then type:

```
Write a PostgreSQL query to count all rows in tbl_customers.
```

The model should return something similar to:

```
</> SQL
```

```
SELECT COUNT(*) FROM tbl_customers;
```

API Test:

`curl http://localhost:11434/api/tags`

Expected:Json response

Step 5: Verify the Ollama API

Keep Ollama installed (you can exit the chat session if it's still running by typing `/bye`).

Now open a new **Command Prompt** window and run:

```
</> Bash  
  
curl http://localhost:11434/api/tags
```

You should receive a *JSON* response containing information about your installed models, including `qwen3:4b` .

Example:

```
</> JSON  
  
{  
  "models": [  
    {  
      "name": "qwen3:4b"  
    }  
  ]  
}
```



Installing other models-

About installing other models

You can install multiple models side-by-side. Ollama only loads a model into memory when you run it.

For example:

 Bash

```
ollama pull qwen3:8b
ollama pull mistral
ollama pull llama3.1
ollama pull gemma3
```

All of these can coexist on your machine.

To see what's installed:

 Bash

```
ollama list
```

To use a specific model:

 Bash

```
ollama run qwen3:4b
```



Step 6 - Setup Supabase

- Create Supabase Project
- Import:
 - customers.csv → tbl_customers
 - transactions.csv → tbl_transactions
- Set Primary Keys:
 - Customer_id
 - Transaction_id

Step 7 - Get Database Credentials

- In Supabase obtain:
- Host
- Database Name
- Username
- Password
- Port
- Note:
- Supabase often uses port 6543
- instead of standard PostgreSQL 5432

Step 8 - Create PostgreSQL Credential in n8n

- Add PostgreSQL Credential
- Enter:
- Host
- Database
- User
- Password
- Port
- Test Connection
- Expected:
- Credential saved successfully

Step 9 - Import Data Analyst Workflow

- Import workflow JSON
- Verify nodes:
 - • AI Agent
 - • DB Schema
 - • Get Table Definition
 - • Run SQL Query
 - • Chat Model

Step 10 - Configure Ollama Credential

- Create Ollama Credential
- DO NOT USE:
- `http://localhost:11434`
- USE:
- `http://127.0.0.1:11434`
- Save Credential

2. n8n is trying IPv6 while Ollama is listening on IPv4

This is very common on Windows.

Instead of:

`http://localhost:11434`

edit the Ollama credential and try:

`http://127.0.0.1:11434`

Save it and test again.

Critical Troubleshooting - IPv4 vs IPv6

- Issue Faced:

Error:

- 'The service refused the connection'
- 'Error fetching options from Ollama Chat Model'

- Root Cause:

- localhost resolved differently
- between Windows/n8n/Ollama

- Fix:

- Replace:

- `http://localhost:11434`

With:

- `http://127.0.0.1:11434`

- Result:

- Qwen3 models appeared immediately

Step 11 - Select Model

- In Ollama Chat Model node:
- Select:
- qwen3:8b
- Alternative:
- qwen3:4b
- Execute Test Step
- Expected:
- Successful response generated

Validation Commands

- Node:
- `node -v`

- NPM:
- `npm -v`

- Ollama Models:
- `ollama list`

- Ollama API:
- `curl http://127.0.0.1:11434/api/tags`

- Start n8n:
- `npx n8n`

Architecture

- User Question
- ↓
- n8n AI Agent
- ↓
- Ollama (Qwen3:8B)
- ↓
- SQL Generation
- ↓
- Supabase PostgreSQL
- ↓
- Query Results

Token Understanding

- Example:
- AI Agent Success
- ~999 Tokens
- Meaning:
- Input + Output text processed
- Cost:
- Local Ollama = ₹0
- n8n uses no AI tokens
- Supabase uses no AI tokens

Local vs Cloud LLM

- Local:
- ✓ No API key
- ✓ No token charges
- ✓ Private
- ✓ Offline
- Cloud:
- ✓ Stronger models
- ✓ Faster
- ✗ API costs
- ✗ Internet required

Advantages

- • No OpenAI/OpenRouter costs
- • Unlimited usage
- • Better privacy
- • Local execution
- • Learn AI agents safely
- • Good for SQL generation
- • Good for automation workflows

Common Issues & Fixes

- Issue: node not recognized
- Fix: Install Node.js and reopen terminal

- Issue: Model not visible
- Fix: Use 127.0.0.1

- Issue: Connection refused
- Fix: Check Ollama running

- Issue: PostgreSQL error
- Fix: Verify host/user/password/port

- Issue: Antivirus blocks access
- Fix: Check firewall/AVG rules

Live Demo Questions

- Show top 10 customers by revenue
- Which customer placed most orders?
- Show monthly sales trend
- List customers with no transactions
- What is total transaction value?

Commands -

node -v

npm -v

ollama --version

ollama list

curl http://127.0.0.1:11434/api/tags

npx n8n